Thermodynamic Dissolution of Operational Simultaneity Across Gravitational Horizons and Galactic Timing Arrays

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Abstract

Debates surrounding the black hole information paradox and horizon complementarity frequently treat the event horizon ($r = r_s$) as an abrupt geometric boundary where quantum mechanics and general relativity collide, forcing seemingly incompatible descriptions between infalling and distant stationary observers. Concurrently, in galactic-scale metrology, Pulsar Timing Arrays (PTAs) correlate millisecond pulsar signals across kiloparsec baselines to reconstruct cosmological gravitational-wave backgrounds. In this capstone paper of Part II, we extend the Coherence-Bounded Operational Locality framework and the Operational Simultaneity Horizon ($\mathcal{L}_{\text{sim}}$) into curved Schwarzschild spacetime and extreme astrophysical baselines. We derive the **Redshift-Modulated Operational Simultaneity Horizon** $\mathcal{L}_{\text{sim}}(r) = \mathcal{L}_{\text{sim},\infty}\sqrt{1 - r_s/r}$ and prove that for any finite proper separation $\Delta\ell > 0$ and non-zero thermal/quantum phase decoherence rate $\Gamma_{\text{tot}} > 0$, active phase synchronization dissolves at a finite *pre-horizon dissolution radius* $r_{\text{diss}} > r_s$ strictly outside the classical event horizon. Within a neutral four-dimensionalist ontology, operational simultaneity dissolves smoothly via thermodynamic signaling decay ($\mathcal{D}_{\text{loc}} \to 0$) rather than via discontinuous geometric pathology. Finally, we demonstrate that kiloparsec Pulsar Timing Arrays operate deep within the $\mathcal{D}_{\text{loc}} \to 0$ regime, proving that galactic-scale simultaneity is structurally restricted to post-selected classical coincidence correlation.

1 Introduction: From Flat-Spacetime Horizons to Curved Metrology

In Papers 9 through 12 of this series, we demonstrated that active real-time phase synchronization across a proper baseline $\Delta\ell$ is bounded by the Operational Simultaneity Horizon $\mathcal{L}_{\text{sim}} = (cN_{\text{eff}}\rho_\tau)/(2\Gamma_{\text{tot}})$, governed by the Spatial Locality Decay Factor $\mathcal{D}_{\text{loc}} = \exp(-\Delta\ell/\mathcal{L}_{\text{sim}})$. While post-selected quantum correlations (S_{CHSH}) remain invariant across spacelike intervals, active operational causal control decays exponentially once $\Delta\ell$ approaches $\mathcal{L}_{\text{sim}}$.

In flat Minkowski spacetime, $\mathcal{L}_{\text{sim}}$ is limited solely by the refractive index N_{eff}, the target locking threshold ρ_τ, and the total phase decoherence rate Γ_{tot}. In strong gravitational fields, however, spacetime curvature introduces gravitational time dilation and coordinate signal delay (Shapiro delay). In this paper, we examine two extreme regimes where gravitational and astrophysical scales govern operational simultaneity:

1. **Strong-Field Gravitational Horizons ($r \to r_s$):** Where gravitational redshift stretches two-way signal round-trip times, collapsing $\mathcal{L}_{\text{sim}}(r) \to 0$ outside the classical event horizon.

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2. **Galactic-Scale Pulsar Timing Arrays ($\Delta\ell \sim$ kpc):** Where astronomical baselines place multi-clock networks permanently inside the $\mathcal{D}_{\rm loc} \to 0$ post-selected regime.

2 Operational Simultaneity in Schwarzschild Spacetime

Consider a static, spherically symmetric Schwarzschild spacetime with line element ($G = c = 1$ in geometric units, restoring c explicitly in operational formulas):

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2, \tag{1}$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius.

Definition 1 (Redshift-Modulated Operational Simultaneity Horizon)**.** *For a stationary local observer at radial coordinate $r > r_s$ maintaining an optical clock network in vacuum ($N_{\rm eff} = 1$) with local proper decoherence rate $\Gamma_{\rm tot}$ and locking threshold ρ_τ, the local proper round-trip signaling speed is modulated by the lapse function $\alpha(r) = \sqrt{1 - r_s/r}$. The **Redshift-Modulated Operational Simultaneity Horizon** $\mathcal{L}_{\rm sim}(r)$ is defined as:*

$$\mathcal{L}_{\rm sim}(r) = \mathcal{L}_{\rm sim,\infty}\sqrt{1 - \frac{r_s}{r}} = \frac{c\rho_\tau}{2\Gamma_{\rm tot}}\sqrt{1 - \frac{r_s}{r}}, \tag{2}$$

where $\mathcal{L}_{\rm sim,\infty} = (c\rho_\tau)/(2\Gamma_{\rm tot})$ denotes the asymptotic flat-spacetime horizon as $r \to \infty$.

Theorem 1 (Pre-Horizon Thermodynamic Dissolution of Simultaneity)**.** *For any two spatially separated clock nodes maintaining a non-zero proper radial separation $\Delta\ell > 0$ and non-zero phase decoherence $\Gamma_{\rm tot} > 0$ ($\mathcal{L}_{\rm sim,\infty} < \infty$), active operational phase synchronization ($\mathcal{D}_{\rm loc} \geq e^{-1}$) fails at a finite critical radius $r_{\rm diss}$ strictly outside the classical event horizon ($r_{\rm diss} > r_s$):*

$$r_{\rm diss} = \frac{r_s}{1 - \left(\Delta\ell/\mathcal{L}_{\rm sim,\infty}\right)^2} > r_s \quad \text{for } 0 < \Delta\ell < \mathcal{L}_{\rm sim,\infty}. \tag{3}$$

Proof. Evaluating the Spatial Locality Decay Factor $\mathcal{D}_{\rm loc}(r) = \exp\left(-\Delta\ell/\mathcal{L}_{\rm sim}(r)\right)$ at radial coordinate $r > r_s$ yields:

$$\mathcal{D}_{\rm loc}(r) = \exp\left(-\frac{\Delta\ell}{\mathcal{L}_{\rm sim,\infty}\sqrt{1 - r_s/r}}\right). \tag{4}$$

Setting the operational horizon threshold $\mathcal{D}_{\rm loc}(r_{\rm diss}) = e^{-1}$ requires $\Delta\ell = \mathcal{L}_{\rm sim,\infty}\sqrt{1 - r_s/r_{\rm diss}}$. Squaring both sides and solving for $r_{\rm diss}$ gives:

$$1 - \frac{r_s}{r_{\rm diss}} = \left(\frac{\Delta\ell}{\mathcal{L}_{\rm sim,\infty}}\right)^2 \implies r_{\rm diss} = \frac{r_s}{1 - \left(\Delta\ell/\mathcal{L}_{\rm sim,\infty}\right)^2}. \tag{5}$$

Since $\Delta\ell > 0$ and $\mathcal{L}_{\rm sim,\infty} < \infty$, the denominator satisfies $0 < 1 - (\Delta\ell/\mathcal{L}_{\rm sim,\infty})^2 < 1$, guaranteeing $r_{\rm diss} > r_s$. As $r \to r_s^+$, $\mathcal{L}_{\rm sim}(r) \to 0$ and $\mathcal{D}_{\rm loc}(r) \to 0$ exponentially. □

Corollary 1 (Smooth Dissolution of Naive Horizon Complementarity)**.** *Because $\mathcal{D}_{\rm loc}(r) \to 0$ smoothly prior to $r = r_s$ (further accelerated near the horizon by Unruh-Hawking thermal noise $\Gamma_{\rm tot}(r) \propto (1 - r_s/r)^{-1/2}$), no joint real-time operational simultaneity frame can span the horizon. Apparent horizon complementarity paradoxes arise from extrapolating naive classical point-locality ($\mathcal{D}_{\rm loc} \equiv 1$) across a boundary where operational synchronization has already thermodynamically dissolved.*

3 Galactic-Scale Metrology: Pulsar Timing Arrays

At the opposite extreme of low curvature but vast spatial separation, Pulsar Timing Arrays (PTAs) monitor ensembles of millisecond pulsars separated by baselines $\Delta\ell \sim 1$ to 10 kpc ($\sim 3 \times 10^{19}$ to 3×10^{20} m).

Even for an ultra-stable millisecond pulsar exhibiting an effective timing/phase decoherence rate $\Gamma_{\text{tot}} \sim 10^{-11}$ s^{-1} and $\rho_\tau = 0.5$, the asymptotic Operational Simultaneity Horizon evaluates to:

$$\mathcal{L}_{\text{sim},\infty} = \frac{(2.9979 \times 10^8 \text{ m/s})(0.5)}{2(10^{-11} \text{ s}^{-1})} \approx 7.5 \times 10^{18} \text{ m} \approx 0.24 \text{ kpc}. \tag{6}$$

Across a typical 5 kpc galactic baseline, $\Delta\ell/\mathcal{L}_{\text{sim},\infty} \approx 20.8$, yielding:

$$\mathcal{D}_{\text{loc}} = \exp(-20.8) \approx 9.3 \times 10^{-10} \to 0. \tag{7}$$

Table 1: Comparison of Operational Simultaneity Regimes Across Laboratory, Astrophysical, and Horizon Scales.

Physical Regime	**Baseline / Radius**	$\mathcal{D}_{\text{loc}}$ **Value**	**Operational Metrology Status**
Local Quantum Core	$\Delta\ell \ll \mathcal{L}_{\text{sim}}$	$\mathcal{D}_{\text{loc}} \to 1$	Active real-time phase-locked synchroniza
Long-Baseline Fiber	$\Delta\ell \sim$ 10–50 km	$\mathcal{D}_{\text{loc}} \in [0.1, 0.7]$	Horizon transition; bifurcation vs. S_{CHSH}
Galactic PTA Network	$\Delta\ell \sim$ 1–10 kpc	$\mathcal{D}_{\text{loc}} \sim 10^{-10} \to 0$	Strictly post-selected classical correlation
Near-Horizon Region	$r \to r_{\text{diss}} > r_s$	$\mathcal{D}_{\text{loc}} \to 0$	Pre-horizon dissolution of simultaneity

Consequently, galactic-scale timing arrays cannot construct an active, real-time cosmic simultaneity hyperplane. Instead, PTA gravitational-wave detection operates strictly in the post-selected classical coincidence regime—structurally analogous to post-selected S_{CHSH} correlation records in Paper 10—reconstructing spacetime strain correlations retrospectively within the four-dimensional block.

4 Philosophical Synthesis: Four-Dimensionalism Without Paradox

Across the entire arc of Papers 9 through 13, a unified philosophical picture emerges within a neutral four-dimensionalist Block Universe ontology:

1. **Ontological Completeness:** All events—whether laboratory optical clock ticks, spacelike-separated Bell measurements, pulsar pulse emissions, or worldlines crossing $r = r_s$—occupy well-defined, non-paradoxical locations on the four-dimensional Lorentzian manifold.

2. **Operational Epistemology:** What observers experience as "simultaneity" or "local causal control" is not a primitive geometric slicing of the manifold, but a thermodynamically bounded operational achievement governed by $\mathcal{D}_{\text{loc}} = \exp(-\Delta\ell/\mathcal{L}_{\text{sim}}(r))$.

3. **Resolution of Paradoxes:** From the Andromeda Paradox (Paper 9) and Bell nonlocality (Paper 10) to non-unitary Quantum Reference Frames (Paper 11), quantum gravity interferometry calibration (Paper 12), and gravitational horizon complementarity (Paper 13), apparent foundational paradoxes dissolve once idealized point-locality ($\mathcal{D}_{\text{loc}} \equiv 1$) is replaced by Coherence-Bounded Operational Locality.

5 Conclusion

By extending the Operational Simultaneity Horizon into curved Schwarzschild spacetime and galactic-scale baselines, we establish that operational simultaneity dissolves smoothly at a finite radius $r_{\text{diss}} > r_s$ outside black hole event horizons and vanishes across kiloparsec Pulsar Timing Arrays. Together with Papers 10 through 12, this completes Part II of the Temporal Coherence & Operational Relativistic Metrology series, providing a continuous mathematical bridge from tabletop optical clocks to relativistic astrophysics.

AI Collaboration & Authorship Transparency Statement

This work is the result of human-AI scholarly collaboration. The core philosophical framework, conceptual architecture, operational definitions, and theoretical direction were conceived and authored by **Luís Pimenta Damásio** (ORCID: 0009-0001-7606-7557). Mathematical formalization, analytical verification, LaTeX typesetting, and structural drafting support were provided by **Gemini** (Google AI) under the direct supervision, review, and editorial authority of the human author.

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